ASTRO-WORLDMODELS · EVALUATION PROGRAM

AstroJEPA Benchmark Report

Where the LeJePA image and spectra backbones stand against AstroCLIP and AION-1 — measured on matched samples, matched heads, and honest metrics.

branch **redshift-regression-eval** 2026-08-12 repo pranavktrpl/Astro-Worldmodels

CROSS-MATCH REDSHIFT R^2
0.551
adapted ViT-L · **AstroCLIP 0.79**

GALAXY10 REDSHIFT R^2 ($Z<0.25$)
0.748
adapted ViT-L · best in repo

GZD-5 MORPHOLOGY ACC
0.761
adapted ViT-L · **AstroCLIP 0.761**

SPECTRA REDSHIFT R^2
0.432
partial collapse · **AstroCLIP 0.98**

PROVABGS STELLAR MASS R^2
0.506
spectra · **AstroCLIP spec 0.88**

01 Key findings

01 **The gap to AION is representational, not a probe artifact.** Reproducing AION's exact 2-layer MLP head gained only +0.6 pts over our linear probe (0.718 → 0.724 on Galaxy10). Head capacity is ruled out as an explanation for the ~15-point morphology gap.

02 **On AstroCLIP's own sample, the honest redshift gap is 0.55 vs 0.79.** Evaluating on their exact DESI-LS × DESI cross-match and 80/20 split (not Galaxy10's bright low-z population) shows the deficit is concentrated in the faint mag 20–21 population our pretraining underrepresents.

03 **Domain adaptation is a free win at both scales.** Ten epochs of continued LeJePA on the 139k cross-match train images: +2.1 R^2 on the cross-match, +0.9–1.4 on Galaxy10, morphology-positive for ViT-S (how-rounded +0.049 at 5.6σ), morphology-neutral for ViT-L — and zero regressions anywhere, multi-

seed verified. The adapted checkpoints are the recommended pair.

04 **The GZD-5 benchmark was 60% broken — now fixed.** Under AstroCLIP's own protocol (softmax-before-CE), 6 of 10 questions collapse to majority-class prediction: frozen validation loss, bit-identical scores across all backbones. A fixed-head protocol (logits-CE + class weights) makes all ten questions discriminative; honest mean F1 is 0.63–0.66, and the S→L scaling advantage concentrates in how-rounded (+0.109, 10σ).

05 **Embedding geometry is the structural weakness.** kNN is consistently our worst head and AstroCLIP's best (their kNN 0.79 beats their MLP): LeJePA embeddings are linearly decodable but metrically loose. This is the direct motivation for the CLIP-style image↔spectrum alignment stage.

06 **The spectra backbone works but is rank-collapsed.** First-ever evaluation: redshift R^2 0.432 vs AstroCLIP's 0.98; PROVABGS properties at a consistent 40–80% of their spectrum encoder. Diagnostics confirm low effective rank; run5 anti-collapse pretraining changes are the response. Notably, even collapsed, it already beats AstroCLIP's *image* encoder on sSFR (0.51 vs 0.42).

02 Redshift — image backbones

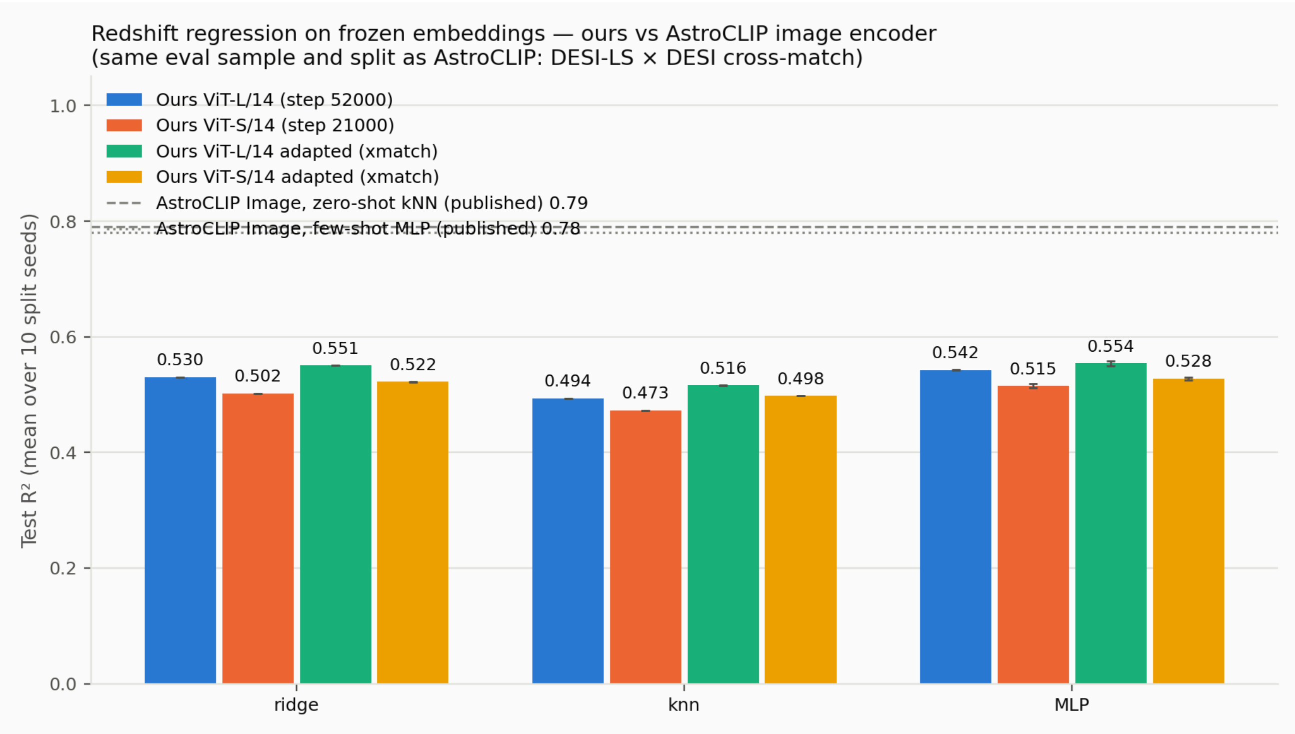
Both eval samples, ridge head (the most stable), baseline vs adapted. The cross-match rows are directly comparable to AstroCLIP's published numbers: same galaxies, same split, same image-only input.

TEST R^2 · RIDGE ON FROZEN EMBEDDINGS · MEAN OVER SEEDS

MODEL	CROSS-MATCH	Δ ADAPT	GALAXY10 (Z<0.25)	Δ ADAPT
ViT-S/14 baseline	0.502		0.674	
ViT-S/14 adapted	0.522	+0.020	0.688	+0.014
ViT-L/14 baseline	0.530		0.739	
ViT-L/14 adapted	**0.551**	+0.021	**0.748**	+0.009
AstroCLIP image, zero-shot kNN	0.79		—	

AION-1-L, photometry+image	0.94*	—

*AION uses photometry inputs and a different cross-matched sample; the AstroCLIP image rows are the like-for-like reference. Galaxy10 unclipped R^2 is tail-dominated (per-seed swings 0.21–0.76) and deliberately not reported.



All heads on the matched sample. Adaptation lifts every bar; the kNN column shows the geometry gap most clearly.

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03 Morphology

Galaxy10 with matched competitor heads removes the head-capacity confound; GZD-5 is the like-for-like AstroCLIP benchmark (their labels, their head), reported under both their protocol and the fixed one.

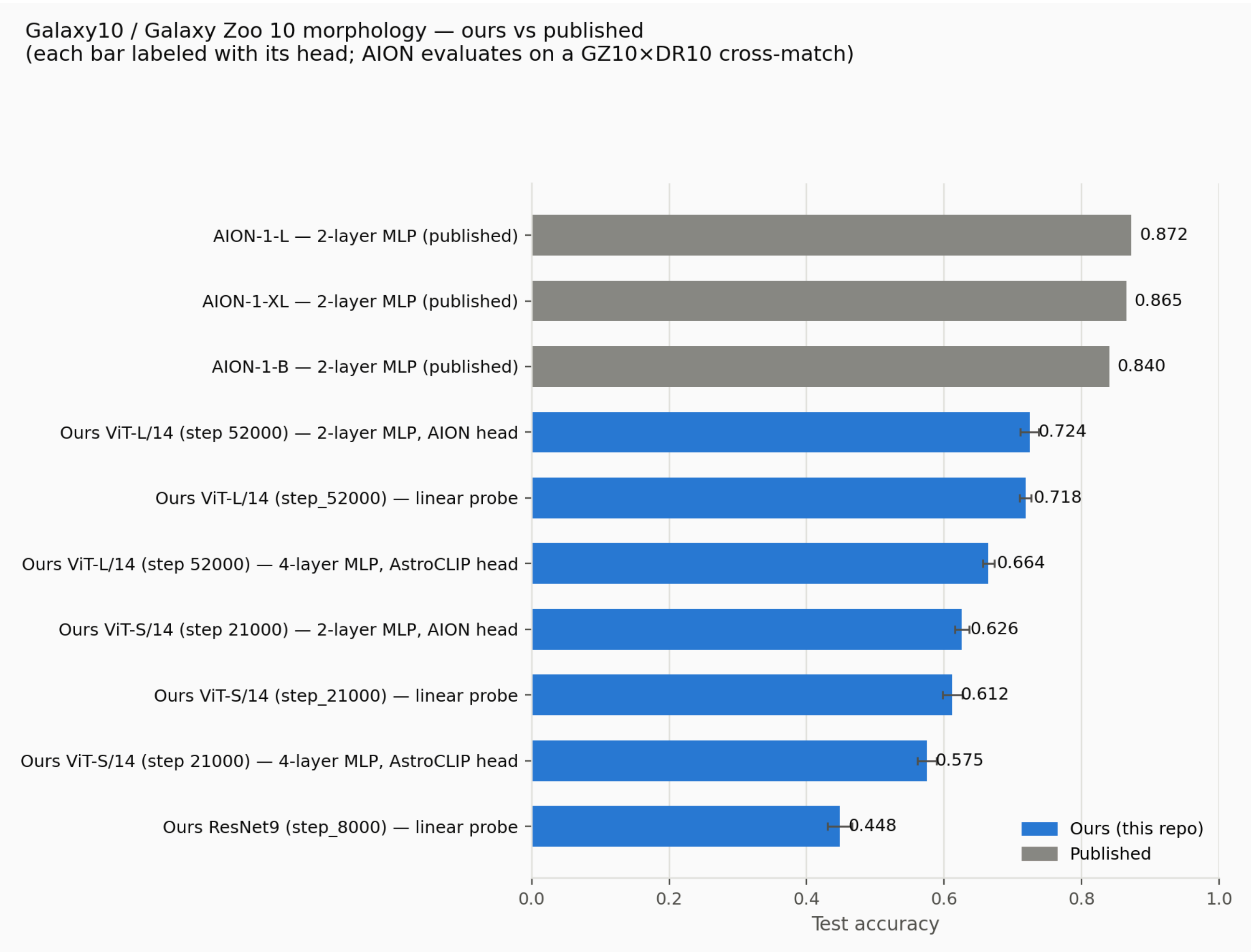
GALAXY10 · TEST ACCURACY · 3 SEEDS

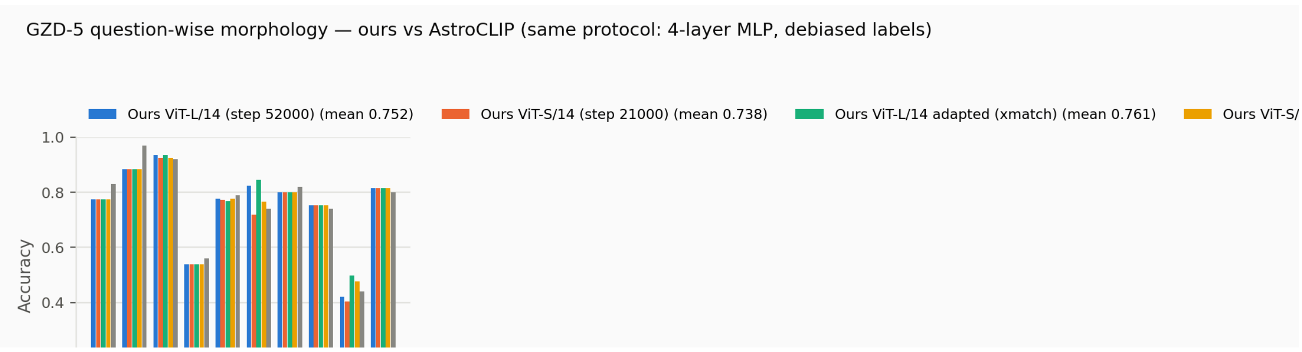
MODEL / HEAD	LINEAR PROBE	2-LAYER MLP (AION HEAD)	4-LAYER M
ViT-S/14	0.612	0.626	
ViT-L/14	0.718	**0.724**	
AION-1-L (published, GZ10×DR10)	—	0.872	

GZD-5 · MEAN OVER 10 QUESTIONS · FIXED HEAD = 5 SEEDS

MODEL	REPRO ACC	REPRO F1	FIXED-HEAD F1 ±Σ
ViT-S/14 baseline	0.738	0.676	0.623 ± 0.015
ViT-S/14 adapted	0.751	0.686	0.633 ± 0.014
ViT-L/14 baseline	0.752	0.689	**0.653 ± 0.016**
ViT-L/14 adapted	**0.761**	0.695	0.655 ± 0.013
AstroCLIP image (published)	0.761	0.743	n/a — protocol collapses

Repro-protocol means are inflated by the six majority-collapse questions and kept only for cross-paper comparability; fixed-head numbers are the honest internal ranking.





Per-question view (repro protocol). The six questions where all bars tie are the majority-collapse artifact — see finding 04.

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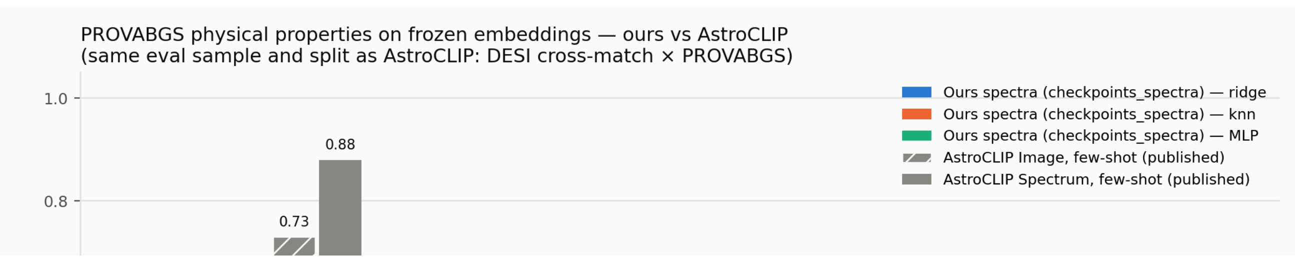
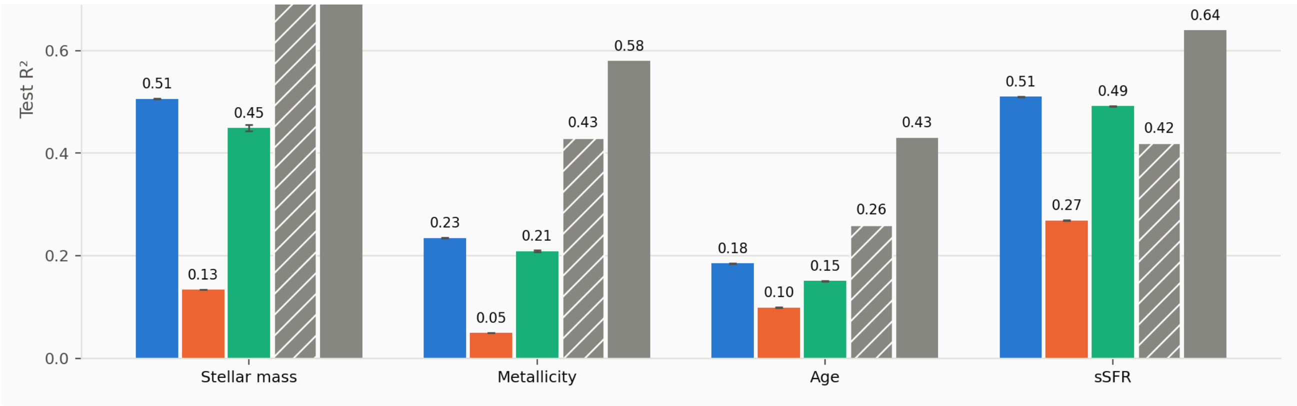
04 Spectra backbone FIRST EVAL · PARTIAL COLLAPSE

Frozen `SpectrumTransformerEncoder` (step 15647) on the same cross-match sample and split: redshift from spectra, and PROVABGS physical properties joined on targetid (89,334 matches). Diagnostics verdict: low effective rank — the embedding carries real spectral physics but a collapsed representation caps what the heads can extract. Run5 anti-collapse pretraining changes are in flight.

SPECTRA EMBEDDINGS · TEST R^2 · RIDGE · 3 SEEDS

TARGET	OURS (SPECTRA)	ASTROCLIP SPECTRUM	ASTROCLIP IMAGE
Redshift	0.432	0.98	0.79
Stellar mass	0.506	0.88	0.73
sSFR	**0.510**	0.64	0.42
Metallicity	0.235	0.58	0.43
Age	0.185	0.43	0.26





Even partially collapsed, the spectra backbone beats AstroCLIP's image encoder on sSFR — but it must approach their spectrum encoder before it can serve as the alignment partner.

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05 Compute accounting — are we undertrained?

Yes, everywhere — and unevenly. Training FLOPs are estimated as $6 \times N_{backbone} \times D$, where D is transformer tokens processed across all ten multi-crop views (2 global 140px + 8 local 56px = 338 tokens/image at patch 14; spectra: 10 × 390 tokens). Sample counts follow from the committed configs.

TRAINING SCALE PER MODEL · FLOPS ≈ 6·N·D

RUN	PARAMS	UNIQUE SAMPLES	SAMPLES SEEN	EPOCHS
ViT-L/14 pretrain	304.4 M	8.44 M	39.9 M	4.7
ViT-S/14 pretrain	22.1 M	8.44 M	16.1 M	1.9
ViT-L adaptation (increment)	304.4 M	0.14 M	1.4 M	10
ViT-S adaptation (increment)	22.1 M	0.14 M	1.4 M	10
Spectra pretrain	86 M	1.13 M	1.0 M	0.9
AstroCLIP image (published)	307 M	76 M	n/r	n/r
AION-1 (published)	B/L/XL tiers	200 M+ obs.	n/r	n/r

Three ratios carry the argument:

✓ **ViT-L is parameter-heavy, data-light:** 27.7k unique samples per million parameters, vs ~248k for AstroCLIP's image encoder — a 9× exposure deficit at similar model size, before counting their extra epochs.

- ✓ **The spectra backbone never completed an epoch:** 0.9 passes over 1.13 M spectra, the most compute-starved model in the program — collapse is unsurprising at this dose, independent of the objective.
- ✓ **Adaptation is extreme-leverage compute:** the ViT-S adaptation cost 0.3% of the ViT-L pretrain budget and bought +2 R^2 on the cross-match plus gains everywhere else.

Assumptions: 4 GPUs for all runs (documented for ViT-L; assumed for ViT-S and spectra), ViT-S on the same bs 192 recipe, projector FLOPs ignored, competitor training compute not published.

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06 Measurement integrity

Every confound raised during the program has been closed:

- ✓ **Head capacity** — competitor heads reproduced exactly (architecture, loss, quirks included).
- ✓ **Sample identity** — evaluation on AstroCLIP's own cross-match file, their split preserved; dr2-RGB preprocessing verified against their code.
- ✓ **Metric honesty** — clipped R^2 for tail-dominated redshift; fixed-head protocol for the GZD-5 collapse; weighted-F1 inflation documented.
- ✓ **Leakage** — train/test targetids verified disjoint; checkpoint and hyperparameter selection on validation only; test touched once per claim.
- ✓ **Seed noise** — 10 seeds (redshift), 5 seeds (fixed-head GZD-5), 3 seeds elsewhere; one headline-grade finding was retracted after failing a 5-seed test.

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07 What to do next

Ordered by expected return per FLOP, grounded in the accounting above:

- → **Finish training the spectra backbone (run5, ≥10 epochs).** It has seen 0.9 epochs; ten epochs cost ~2×10^{19} FLOPs — the same as the ViT-L run already spent — and redshift-from-spectra has the largest headroom in the program (0.43 → 0.98 ceiling). Gates the alignment stage.

→ **Spend the next image budget on data, not parameters.** ViT-S on the 76 M-image Stein/ssl-legacysurvey corpus for ~4 epochs costs $\sim 1.4 \times 10^{19}$ FLOPs — *half* the ViT-L budget — while multiplying unique data 9× and fixing the Chinchilla imbalance the repo's own notes flagged. ViT-L at 2.9× the score-per-FLOP disadvantage is the wrong place to add compute until data catches up.

→ **Run the adaptation dose-response** (lr 2e-4 × 30 epochs, ~0.9% of ViT-L budget): if the +2 R^2 keeps scaling, the big pretrain above should be faint-population-heavy.

→ **CLIP alignment on the 139k cross-match pairs** once run5 probes well — trains on cached embeddings in minutes; the direct fix for the kNN geometry gap (finding 05).

→ **Image-side PROVABGS probe** — cached embeddings + existing targetid join; fills the last empty comparison column at zero GPU cost.

`Generated from committed metrics on redshift-regression-eval · reproduce with python Evals/competitor_comparison/compare_competitors.py · published baselines transcribed in competitor_baselines.json`